Worksheet 6A: Statistics and Probability

What This Worksheet Is About

Statistics and probability are the mathematical foundation of machine learning. Every time an AI makes a prediction, classifies an image, or recommends a video, it's using these concepts.

This worksheet connects the basics you've learned to real applications — from training neural networks to analyzing scientific data.

Part A: Measures of Central Tendency — Mean, Median, Mode (18 problems)

The Formulas

Mean (Average): $\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$

Median: The middle value when data is sorted (or average of two middle values if n is even)

Mode: The most frequent value

Calculate the mean, median, and mode:

1. Test scores: 78, 82, 85, 85, 88, 90, 92

2. Response times (ms): 120, 135, 140, 145, 145, 150, 180

3. Daily website visitors: 1200, 1350, 1400, 1450, 1500, 1550, 8500

- (a) Calculate all three measures.
- (b) Which measure is most affected by the outlier (8500)?
- (c) Which measure best represents a "typical" day?

** AI Connection: Training Data Quality**

4. A dataset of house prices (in €1000s) contains: 250, 275, 290, 310, 325, 340, 2500

The last value is a mansion that got mixed in with regular houses.

- (a) Calculate the mean with and without the outlier.
- (b) Calculate the median with and without the outlier.
- (c) Why might a data scientist remove outliers before training a model?

5. An image classification dataset has these label counts:

Category	Count
Cat	5000
Dog	4800
Bird	1200
Fish	500

- (a) What is the mean number of images per category?
- (b) What is the median?
- (c) What is the mode?
- (d) This is called an "imbalanced dataset." Why might this be a problem for training?

** Science Connection: Experimental Measurements**

6. A physicist measures the speed of light in 5 trials (in units of 10^8 m/s):

2.998, 2.995, 3.001, 2.997, 2.999

- (a) Calculate the mean.
- (b) The accepted value is 2.998. How close is the experimental mean?

7. A chemist measures the pH of a solution 7 times:

7.02, 6.98, 7.01, 6.99, 7.00, 7.01, 7.00

- (a) Find mean, median, and mode.
- (b) Are they all similar? What does this suggest about the measurement consistency?

** Game Analytics**

8. Player scores in a mobile game level: 1250, 1480, 1520, 1550, 1600, 1650, 1700, 1800, 1850, 2400

- (a) Find the mean and median.
- (b) If the game designer wants to set the "3-star threshold" at a score that about half of players achieve, should they use the mean or median?

9. Daily active users for an app over two weeks:

Week 1: 10000, 12000, 15000, 14000, 13000, 8000, 7000 Week 2: 11000, 13000, 16000, 15000, 14000, 9000, 8000

- (a) Calculate the mean for each week.
- (b) Is the app growing? By what percentage?

Weighted Mean:

10. A student's grade is calculated as: - Homework: 25% weight, score 85 - Midterm: 30% weight, score 78
- Final: 45% weight, score 88

Calculate the weighted mean (final grade):

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

11. In a neural network, outputs from three layers are combined with weights: - Layer 1: output 0.7, weight 0.5 - Layer 2: output 0.3, weight 0.3 - Layer 3: output 0.8, weight 0.2

Calculate the weighted average output.

Part B: Measures of Spread — Range, Variance, Standard Deviation (16 problems)

The Formulas

Range: $R = x_{max} - x_{min}$

Variance: $\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{n}$

Standard Deviation: $\sigma = \sqrt{\sigma^2}$

Note: For samples, divide by $(n-1)$ *instead of* n *— this is called Bessel's correction.*

Calculate the range, variance, and standard deviation:

12. Data: 4, 6, 8, 10, 12

- (a) Find the mean.
- (b) Find $(x_i - \bar{x})$ for each value.
- (c) Find $(x_i - \bar{x})^2$ for each value.
- (d) Find the variance (divide by $n = 5$).
- (e) Find the standard deviation.

13. Data: 7, 7, 8, 8, 8, 9, 9

14. Data: 2, 5, 8, 11, 14

** AI Connection: Model Consistency**

15. Two machine learning models are tested 5 times each on accuracy (%):

Model A: 85, 87, 84, 86, 88 Model B: 70, 95, 75, 90, 80

- (a) Calculate the mean accuracy for each.
- (b) Calculate the standard deviation for each.

- (c) Which model is more consistent?
- (d) In production, would you prefer a consistent model or one with higher peaks?

16. A chatbot's response times (seconds) over 6 queries:

Version 1.0: 1.2, 1.3, 1.1, 1.4, 1.2, 1.3 Version 2.0: 0.8, 2.1, 0.7, 1.8, 0.9, 1.7

- (a) Which version has lower mean response time?
- (b) Which version is more consistent (lower standard deviation)?
- (c) Users hate unpredictable wait times. Which version would you deploy?

** Science Connection: Measurement Precision**

17. Two instruments measure the same voltage (in volts):

Instrument A: 5.02, 4.98, 5.01, 4.99, 5.00 Instrument B: 5.10, 4.90, 5.15, 4.85, 5.00

- (a) Both have the same mean. Verify this.
- (b) Calculate the standard deviation for each.
- (c) Which instrument is more precise?

18. In manufacturing, parts must be within tolerance. A machine produces bolts with target diameter 10.00 mm. A sample of 5 bolts measures:

10.02, 9.98, 10.01, 9.99, 10.00

- (a) Find the mean and standard deviation.
- (b) If tolerance is ± 0.05 mm (so 9.95 to 10.05 is acceptable), are all bolts within tolerance?
- (c) If the standard deviation were 0.03 mm instead, what percentage of bolts would you expect to be outside tolerance? (Assume normal distribution where ~68% are within 1 SD of mean)

** Understanding Standard Deviation:**

19. For a normal distribution: - ~68% of data falls within 1 standard deviation of the mean - ~95% falls within 2 standard deviations - ~99.7% falls within 3 standard deviations

IQ scores have mean 100 and standard deviation 15.

- (a) What range contains about 68% of people?
- (b) What range contains about 95% of people?
- (c) What percentage of people have IQ above 130?

20. Human height (adult males) has mean 175 cm and SD 7 cm.

- (a) What range contains the middle 68%?
- (b) How tall would someone need to be to be in the top 2.5%?
- (c) A basketball team requires players at least 190 cm tall. Approximately what percentage of men qualify?

Part C: Basic Probability (20 problems)

The Fundamentals

$$P(\text{event}) = \frac{\text{Number of favorable outcomes}}{\text{Total possible outcomes}}$$

Key Rules: - $0 \leq P(A) \leq 1$ - $P(\text{certain event}) = 1$ - $P(\text{impossible event}) = 0$ - $P(\text{not } A) = 1 - P(A)$

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21. A standard deck has 52 cards (4 suits $\times$ 13 values). Find: - (a) P(drawing a heart) - (b) P(drawing an ace) - (c) P(drawing the ace of hearts) - (d) P(not drawing a face card) (face cards = J, Q, K)

22. A bag contains 5 red, 3 blue, and 2 green marbles. Find: - (a) P(red) - (b) P(not red) - (c) P(red or blue)

** AI Connection: Classification Probabilities**

23. A spam filter classifies 1000 emails: - 800 are correctly identified (700 real emails marked "not spam", 100 spam marked "spam") - 200 are incorrectly identified (150 real emails marked "spam", 50 spam marked "not spam")

- (a) What is P(correct classification)?
- (b) What is P(false positive) — real email marked as spam?
- (c) What is P(false negative) — spam marked as real?
- (d) Which error is worse for the user?

24. An image classifier sees 500 images: - 200 cats (correctly identifies 180) - 200 dogs (correctly identifies 170) - 100 birds (correctly identifies 80)

- (a) What is $P(\text{correct} \mid \text{cat})$ — probability of correct given it's a cat?
- (b) What is $P(\text{correct} \mid \text{dog})$?
- (c) What is $P(\text{correct} \mid \text{bird})$?
- (d) What is the overall accuracy?

** Science Connection: Medical Testing**

25. A disease affects 1% of the population. A test for the disease: - Correctly identifies 95% of sick people (sensitivity) - Correctly identifies 90% of healthy people (specificity)

In a population of 10,000 people: - (a) How many are actually sick? Healthy? - (b) Of the sick, how many test positive? - (c) Of the healthy, how many test positive (false positives)? - (d) If someone tests positive, what's the probability they're actually sick?

This is the famous "base rate fallacy" — the answer may surprise you!

** Game Probability**

26. A loot box in a game contains: - Common item: 70% chance - Rare item: 20% chance - Epic item: 8% chance - Legendary item: 2% chance

- (a) What is $P(\text{rare or better})$?
- (b) If you open 3 boxes, what's the probability of getting no legendary items?
- (c) A player opens 50 boxes and gets no legendary items. How "unlucky" is this? Calculate $P(\text{no legendary in 50 boxes}) = 0.98^{50}$

27. In a card game, you need to draw 2 specific cards from a 20-card deck: - (a) What is $P(\text{first card is correct})$? - (b) Given the first card was correct, what is $P(\text{second card is correct})$? - (c) What is $P(\text{both cards correct})$?

Addition Rule (OR)

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

For mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B)$

28. Rolling a die: - (a) $P(\text{even or greater than 4})$ - (b) Are "even" and "greater than 4" mutually exclusive?

29. Drawing from a deck: - (a) $P(\text{heart or face card})$ - (b) Why do we subtract $P(\text{heart and face card})$?

Multiplication Rule (AND)

Independent events: $P(A \text{ and } B) = P(A) \times P(B)$

Dependent events: $P(A \text{ and } B) = P(A) \times P(B|A)$

30. Flipping a coin and rolling a die: - (a) $P(\text{heads and 6})$ - (b) $P(\text{heads and even})$

31. Drawing two cards without replacement: - (a) $P(\text{both aces})$ - (b) $P(\text{both hearts})$

** AI Connection: Independent Events in ML**

32. A neural network has 3 layers, each with 95% chance of working correctly (independently).

- (a) What is $P(\text{all layers work})$?
- (b) What is $P(\text{at least one layer fails})$?
- (c) If the network has 10 layers, each 99% reliable, what is $P(\text{all work})$?

33. A self-driving car uses 4 independent sensors: - Camera: 99% reliable - LIDAR: 98% reliable - Radar: 97% reliable - Ultrasonic: 95% reliable

- (a) What is $P(\text{all sensors work})$?
- (b) The car needs at least one sensor working. What is $P(\text{at least one works})$?

Part D: Counting Principles (16 problems)

Fundamental Counting Principle

If task 1 can be done in m ways and task 2 can be done in n ways, then both tasks can be done in $m \times n$ ways.

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34. A password has 4 characters: - First character: letter (26 options) - Second: digit (10 options) - Third: letter (26 options) - Fourth: digit (10 options)

How many possible passwords are there?

35. A restaurant offers: - 4 appetizers - 6 main courses - 3 desserts

How many different 3-course meals are possible?

Permutations (Order Matters)

$$P(n, r) = \frac{n!}{(n-r)!}$$

Number of ways to arrange r items from n items.

36. How many ways can 5 people line up for a photo?

37. How many 3-letter "words" (any letters, no repeats) can be made from A, B, C, D, E?

38. In a race with 8 runners, how many ways can gold, silver, and bronze be awarded?

Combinations (Order Doesn't Matter)

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Number of ways to choose r items from n items.

39. How many ways can you choose 3 students from a class of 10 to form a committee?

40. How many 5-card poker hands are possible from a 52-card deck?

41. A lottery requires choosing 6 numbers from 1-49. How many possible tickets?

** AI Connection: Feature Selection**

42. A data scientist has 10 features (variables) and wants to train models using different subsets:

- (a) How many ways to choose exactly 3 features?
- (b) How many ways to choose exactly 5 features?
- (c) How many total subsets of features are there (including empty set)? Hint: Each feature is either in or out = 2^{10}

43. A neural network architect is designing a network with: - Choice of 3 activation functions - Choice of 4 optimization algorithms - Number of hidden layers: 1, 2, 3, or 4 - Learning rate: 5 options

How many different configurations are possible?

** Science Connection: Genetics**

44. DNA has 4 bases: A, T, G, C. A codon is a sequence of 3 bases.

- (a) How many different codons are possible?
- (b) There are 20 amino acids. Why does this work out nicely?

45. In a genetics experiment, a researcher needs to test all pairwise combinations of 8 genes. - (a) How many pairs must be tested? - (b) If each test takes 2 hours, how long will the full experiment take?

Pascal's Triangle Connection

46. Row 5 of Pascal's Triangle is: 1, 5, 10, 10, 5, 1

These are $\binom{5}{0}, \binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4}, \binom{5}{5}$

Verify: $\binom{5}{2} = 10$ using the formula.

47. The sum of row n of Pascal's Triangle is 2^n.

- (a) Verify for row 5: $1 + 5 + 10 + 10 + 5 + 1 = ?$
- (b) This represents all possible subsets of a 5-element set. Why?

Part E: Putting It Together — ML Applications (10 problems)

** Accuracy, Precision, Recall**

48. A model classifying tumors as "malignant" or "benign" produces:

	Predicted Malignant	Predicted Benign
Actually Malignant	85 (True Positive)	15 (False Negative)
Actually Benign	20 (False Positive)	880 (True Negative)

Calculate: - (a) **Accuracy** = (TP + TN) / Total - (b) **Precision** = TP / (TP + FP) — "Of predicted positives, how many are correct?" - (c) **Recall** = TP / (TP + FN) — "Of actual positives, how many did we catch?" - (d) In cancer detection, is it worse to have low precision or low recall?

49. Another model has: - Accuracy: 95% - Precision: 70% - Recall: 99%

What does this tell you about the model's behavior?

** Expected Value**

50. Expected value: $E[X] = \sum x_i \cdot P(x_i)$

A game costs €5 to play. You roll a die: - Roll 6: win €20 - Roll 5: win €10 - Roll anything else: win €0

- (a) Calculate the expected winnings.
- (b) What is the expected profit (winnings minus cost)?
- (c) Is this a fair game?

51. A trading algorithm has: - 60% chance of €100 profit - 25% chance of breaking even - 15% chance of €200 loss

- (a) Calculate the expected value per trade.
- (b) Over 100 trades, what is the expected total profit?

** Bayes' Theorem Preview**

52. Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

A factory has two machines: - Machine A produces 60% of items, with 2% defect rate - Machine B produces 40% of items, with 5% defect rate

An item is selected at random and found to be defective. What is the probability it came from Machine A?

- (a) Find P(defective) = P(def|A)P(A) + P(def|B)P(B)
- (b) Apply Bayes' Theorem

** Scientific Sampling**

53. A biologist wants to estimate the number of fish in a lake using capture-recapture: - Day 1: Capture 100 fish, tag them, release - Day 2: Capture 80 fish, find 10 are tagged

Using the proportion: $\frac{\text{tagged in sample}}{\text{sample size}} \approx \frac{\text{total tagged}}{\text{population}}$

Estimate the total fish population.

** Cross-Validation**

54. In 5-fold cross-validation, a model's accuracy on each fold is:

Fold 1: 88%, Fold 2: 85%, Fold 3: 90%, Fold 4: 87%, Fold 5: 85%

- (a) What is the mean accuracy?
- (b) What is the standard deviation?
- (c) Report the result as "mean ± SD"

** A/B Testing**

55. A website runs an A/B test on a new checkout button: - Version A (old): 1000 visitors, 50 purchases (5% conversion) - Version B (new): 1000 visitors, 65 purchases (6.5% conversion)

- (a) What is the relative improvement of B over A?
- (b) Is this difference "real" or could it be random chance? (This requires statistical tests we'll learn later, but intuitively, is 15 extra purchases out of 1000 a big deal?)
